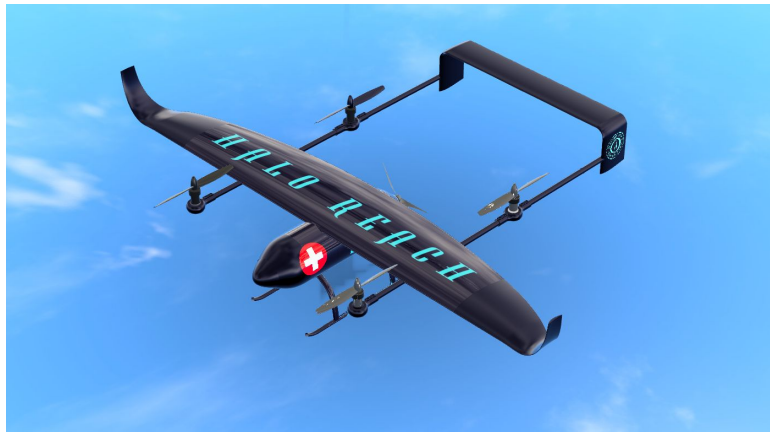

HALO REACH: Wing Design and Computational Propeller Analysis

UAS for Disaster Relief - AE3 2024



Student: Matthew Homa

CID: 02038403

Supervisor: Dr Kostas Steiros

Department: Department of Aeronautics

Course: MEng Aeronautical Engineering

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Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ
U.K.



Table 1. People involved in HALO REACH - GDP

Team code	Team	Names	CID
FATT01	Team Leads	Moreno, Patricia	01857449
		Rashid, Hasan H	02020313
FATT02	Aerodynamics	Alam, Shahriar	02037395
		Dirirsa, Naboth	02026211
		Lawlor, Max F	02026199
		Singh, Parasdeep	02099678
		Homa, Matthew M	02038403
FATT03	Structures	Leng, Jiaxun (Aaron)	02015060
		Wang, Zhiyan	01911209
		Liu, Stephen	02015258
		Chauhan, Shivani	02096006
FATT04	Systems and propulsion	Lin, Yongkai	02029838
		Chursuk, Tharm	02048996
		Wankhade, Divij P	02027906
		Aggarwal, Ishaan K	02030969
		Halim, George	02059571
FATT05	Flight dynamics	Baikady, Thilak S	02016794
		Scott-Brown, Ruairidh	01734080
		Allen, Harry	02028116
		Tang, Shun S	02023280
		Lee, Jeongheon (Edward)	01754474
FATT06	Add-ons	Rosenthal, Adam T	02027024
		Ramteke, Shreya	02079050
		Nakhla, Andrew	02044646

Abstract

The first objective of this study was to design the primary lifting surface for a disaster relief UAV (unmanned aerial vehicle). The fixed wing configuration of the aircraft meant a conventional wing could be designed, and verified, using analytical and computational methods. This resulted in a near-elliptically shaped wing being selected, as this gave the lowest error from an optimal elliptical distribution. The second objective was to use CFD (computational fluid dynamics) to determine, and validate, the thrust and torque of the pusher propeller (the propeller used during the cruise phase of flight) and to compare to BEM (blade element momentum) results used for the initial design. The results showed a very good torque correlation, however the thrust values were consistently lower than the BEM prediction. The reasoning behind this was explored. Positioning the fuselage in front of the propeller improved the propulsive efficiency significantly. This was an exciting discovery as it meant that efficiency could be improved further by simply optimising the fuselage shape as a propeller inlet.

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To avoid repetition, we have endeavoured to keep each report as original as possible. However this meant that some members of the team have been unable to present some of their work, due to tasks being completed by multiple people. In order to try and give full credit to those who worked on topics not covered in their individual reports, a breakdown of work can be found in table 2. As some sections are highly interdependent, please refer to the relevant report for further information. We have tried to ensure that the individual work can be presented standalone however, information from other reports may add value.

Topic	People Involved	Person Presenting
Initial Concepts and Concept Selection	All	Parasdeep
Constraint Diagram	Max	Max
Wing Aerofoil Selection	Max	Max
Main Wing Design	Matthew and Max	Matthew
Analytical Drag Estimates	Max	Max
Winglet Design	Naboth	Naboth
Adkins and Leibeck Code	Parasdeep, Naboth, Shahriar	Parasdeep
Propellor Initial Thrust/Power analysis	Parasdeep	Parasdeep
Propeller Design	Parasdeep	Parasdeep
Propeller performance analysis	Parasdeep	Parasdeep
Rotor Initial Thrust/Power analysis	Shahriar	Shahriar
Rotor Design	Shahriar	Shahriar
Rotor Performance Analysis	Shahriar	Shahriar
CFD Setup and Validation for 2D Aerofoil	Matthew, Max	Max
CFD Setup and Validation For Stock Rotor	Matthew, Max	Matthew
Rotor CFD	Shahriar, Matthew	Shahriar
Propeller CFD	Matthew	Matthew
Full Wing CFD	Naboth	Naboth
Full Aircraft CFD	Max	Max
Blanketing Affects of Tailplane	Naboth	Naboth
Tailplane CFD	Naboth	Naboth

Table 2. Separation of Work

Nomenclature

α_r	Angle of Attack at Wing Root (deg)
α_t	Angle of Attack at Wing tip (deg)
η_p	Propulsive Efficiency
λ	Wing Taper Ratio
ω	Vorticity (s^{-1})
a	Wing Lift Curve Slope (rad^{-1})
a_0	Aerofoil Lift Curve Slope (rad^{-1})
A_n	Collocation Coefficients
AR	Aspect Ratio
b	Wing Span (m)
c	Wing Chord (m)
c_0	Wing Chord at Wing Root (m)
C_d	Drag Coefficient
c_h	Horizontal Tailplane Chord (m)
C_l	Lift Coefficient
e_1	Anderson Efficiency Factor
N	Normal force (N)
N_ω	Non Dimensional Vorticity
P	Power (W)
Q	Torque (Nm)
s	Wing Semi-Span (m)
S_{ref}	Wing Area (m^2)
W	Weight (N)
BEM	Blade Element Momentum
CFD	Computational Fluid Dynamics
GTOW	Gross Takeoff Weight
HALO REACH	Highly Aerodynamic Light Operation Remote Emergency Aid & Crisis Help
RPM	Revolutions Per Minute
UAS	Uncrewed Aerial System
UAV	Unmanned Aerial Vehicle
VTOL	Vertical Take-Off and Landing

1 Introduction and Motivations

The demand for unmanned aerial vehicles (UAVs) has exploded over recent years, and the list of potential applications of this technology is ever growing. Pre-existing long range, high endurance and large payload capacity UAVs seem to pay little attention to the optimal aerodynamic configuration; by utilising off-the-shelf propellers or inefficient wing designs, which focus on aesthetics over performance. The primary objective of this report is to give the aerodynamic design the consideration that it deserves, with the aim to significantly reduce fuel consumption of this UAV, which is essential from an operational costs perspective, while also being a critical environmental consideration.

Part 1 of this report concerns the wing design of the UAV. Analytical methods, such as collocation method and elliptic loading were used to determine a few viable wing designs. Then, XFLR5 (which is an inviscid panel code solver) was used to verify, and refine the design. Eventually it was decided to choose an elliptically shaped wing, as this removed the requirement for twist which would be detrimental to the overall performance, as the wing would be operating at inefficient angles of attack. Winglets were then designed and integrated into the main wing, however this will be covered in Naboth's report [1]. Full viscous analysis was then completed using computational fluid dynamics (CFD), which was used to obtain accurate lift and drag characteristics of the wing and whole UAV, as shown in Max's report [2].

Part 2 discusses the setup, validation and results of CFD simulations on the pusher propeller (propeller used during the cruise phase of the flight), using Star CCM+. This simulation was verified against experimental data of an APC 10x7 propeller, and further validated with a mesh convergence study. These results were then compared to a BEM model with a 3D correction factor using Q Blade. The BEM method is discussed in detail in Parasdeep's report [3]. The results showed a good correlation with BEM for torque, but less so for thrust. These differences were explored, and the reasons behind the differences were discussed. A drop in density in the wake, and turbulence effects were the primary suspects. The CFD simulation was also used to show the significant benefits of the pusher configuration to η_p , by including the fuselage in the simulation. The reasons for this improvement were explored, and suggestions for the next steps were made. It was ultimately determined that more time was needed to perform additional simulations. Then it would be possible to determine how changing airspeed, and fuselage geometry, would impact η_p , allowing an optimal fuselage shape to be design. It would also enable a better understanding of the physics of this phenomenon.

Part I

Wing Design

2 Initial Considerations and Calculations

Firstly, it was deemed necessary to determine the area of the lifting surface. This was done by first fixing the aspect ratio. A high aspect ratio was selected as this is more aerodynamically efficient, however it still needed to be a structurally viable design. It was decided that to set this value so that $AR = 8.5$. This was based on suggested values from Raymer [4], and discussions with the structures sub-team. Using the requirements in the mission brief, S_{ref} and $C_{l_{cruise}}$ were determined using equations 1 and 2. The results, along with mission requirements, are shown in table 3.

$$S_{ref} = b^2/AR \quad (1)$$

$$C_{l_{cruise}} = 2W/\rho V_{cruise}^2 S_{ref} \quad (2)$$

As $M < 0.3$ for the cruise, it was deemed sensible to design the wing from an incompressible aerodynamics perspective, neglect any compressibility effects, and have an unswept wing. The aerofoil selection process is shown in Max's report [2], and this resulted in a NACA 6409 being chosen as it had the optimal L/D in both cruise and loiter stages. The desired spanwise lift distribution is elliptical,

as this profile has the lowest induced drag [5]. Therefore lifting line theory was implemented to obtain this shape, and it was refined and improved using a panel code solver in XFLR5. Although these methods neglect viscous effects, it has been proven that they still accurately calculate the spanwise loading, meaning that an optimal design can be obtained without the use of CFD [6].

Parameter	Value
b (m)	6
S_{ref} (m ²)	4.23
V_{cruise} (m/s)	41.12
W (N)	1570
$C_{lcruise}$	0.375

Table 3. Initial design parameters

3 Collocation Method

The Collocation Method is an application of lifting line theory. It is used to determine the lift and drag coefficients of a wing from the sum of the converging series shown in equations 3 and 4, by calculating the A_n terms, as shown below [7].

$$\mu\alpha\sin(\theta) = \sum_{n=1}^{\infty} A_n \sin(n\theta)(n\mu + \sin(\theta)) \quad (3)$$

$$\mu(\theta) = \frac{a_0(\theta)c(\theta)}{8s} \quad (4)$$

$$C_l = \pi AR A_1 \quad (5)$$

$$C_d = \pi AR (A_1^2 + 3A_3^2 + 5A_5^2 + \dots) \quad (6)$$

In these equations the first three collocation terms were considered, as beyond this, they were so insignificant that they had little impact, yet increased computational cost significantly. It is worth noting that these equations change the frame of reference such that $y = s\cos(\theta)$. The error from an elliptical distribution is calculated as the absolute value of the ratio of the subsequent A terms over A_1 [7].

These equations were solved for a variety of chord and twist distributions. The collocation code, and a graph of results, (shown in appendix A) varies λ , α_r and α_t simultaneously to find out which linear chord and twist distributions that were near elliptical for a given α_t . A linear taper ratio along the span is advantageous from a structural design perspective which is why the use of the collocation method was explored. A brief explanation of the code is highlighted as follows:

- For a fixed α_t , calculate the wing's chord and twist distributions for the tested λ and α_r , assuming linear distributions.
- Solve the collocation code, shown in equations 3 and 4, for the first three collocation coefficients, and obtain L/D and the error from an elliptical distribution.
- For each λ tested, iterate through the results to see which α_r gave the lowest error.
- Run this again for the remaining α_t values, and plot all the lines, allowing three variables to be simultaneously varied.

Unfortunately, the required twist distributions meant that the aerofoil would have operated at very inefficient angles of attack at some stages along the span (it usually changed a lot), hence it was deemed unfeasible to enforce an elliptical lift distribution with twist. The calculated error also only contained the first three collocation terms. Although the remaining terms are very small, they would

have still contributed to the error. XFLR5 was also used to further test some of the potential designs, however none of the designs performed as well as this code predicted (or as well as an elliptically shaped wing, see section 4); probably due to using only three collocation points.

4 Elliptic Wing Geometry

After linear taper was trialled, it was decided to design an elliptically shaped wing to obtain the desired distribution of lift. This method was simpler to analyse and ensured that the aerofoil would always be at an efficient angle of attack, with the small downside of slightly increased structural complexity.

The most famous case of an elliptical wing is the Spitfire. However, although less obvious, elliptical wing designs are still utilised commonly in new aircraft. Some examples are shown in figure 3.

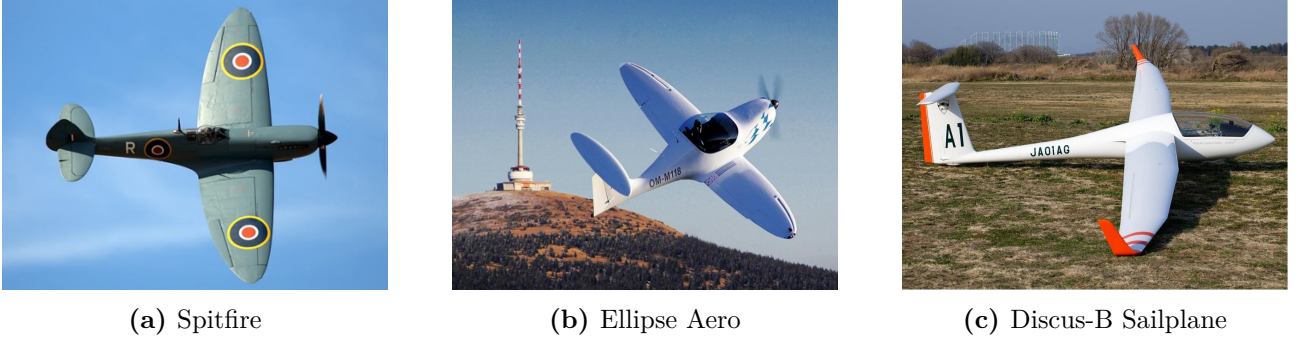


Figure 3. Examples of some elliptical, or semi elliptical, wing designs.

As can be seen in figure 3, there are many different elliptical wing designs. The Discus-B utilises a semi-elliptical lift distribution, which is where all the curvature is on the leading edge of the wing. It was decided to take inspiration from this design and trial different wing designs for elliptical, and semi-elliptical, wings (results shows 4). The formula for an elliptical wing's chord is as follows.

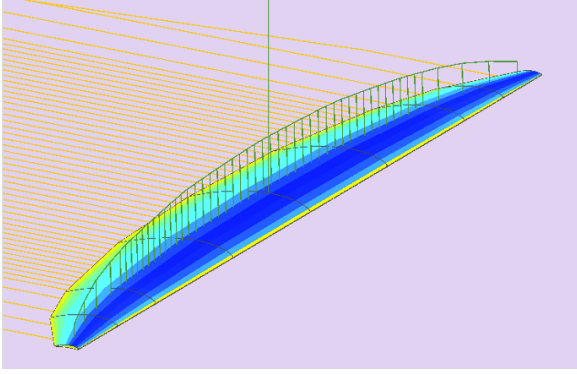
$$c = c_0 \sqrt{1 - \left(\frac{y}{s}\right)^2} \quad (7)$$

It was essential that the wing chord was discretised. A strict requirement from the structures team was to have linear taper between different points along the chord. This makes it easier for them to analyse. The final results obtained an almost elliptical distribution while splitting the wing into 5 sections (with a finer discretisation being used towards the wing tip). The design also allowed for a winglet to be subsequently integrated onto the this wing design. Table 4 below shows the results for the different designs, with the error being calculated as the mean percentage distance from the elliptical distribution.

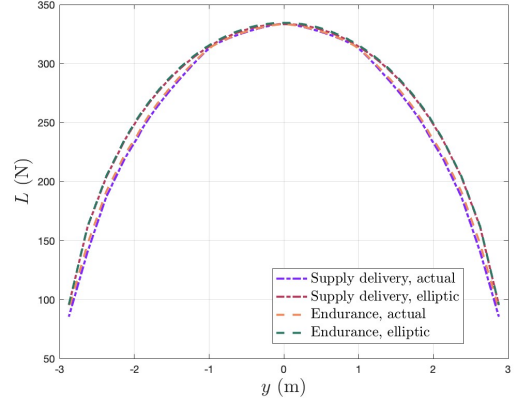
	Full Elliptical	Semi Elliptical (LE straight)	Semi Elliptical (TE straight)
Error (%)	2.97	2.61	3.22

Table 4. Mean error from a completely elliptical distribution

The main reason for this error is because of the requirement to discretise the chord along the span, as this meant that the shape was not perfect. However, the results are very close (especially in the case of a straight leading edge), as shown in figure 4b. Many thanks to Max for performing the simulations of the wing in XFLR.



(a) XFLR wing simulation



(b) Comparison of elliptical to true lift distributions

Figure 4. Comparison of panel code and ideal lift distributions

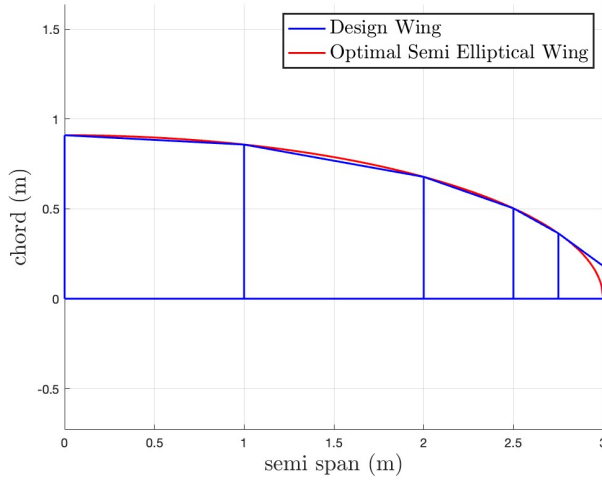


Figure 5. Final wing layout

position (m)	Chord (m)
0	0.910
1	0.858
2	0.678
2.5	0.501
2.75	0.364
3	0.182

Table 5. Wing geometry

Five discretisations was deemed sufficient to ensure it was possible for the wing to be analysed structurally, while also accurately representing an elliptical lift distribution (as proven in table 4). A coarser discretisation meant that the wing did not represent the desired lift distribution well. It is worth noting that this design is excluding winglets. The wing tip was eventually adapted to enable winglets to be added, as covered in Naboth's report [1].

The reason why a wing with a straight leading edge is more efficient is due to a lack of flow outwash effects. In the other two cases in table 4, the leading edge is effectively swept to account for the decreasing chord along the span. This means that some of the free stream air will flow outwards towards the wingtip. This causes losses because free stream air, that should be used to generate lift, is flowing outwards, causing a lift concentration at the tip [7].

4.1 Wing Performance Estimation

Once the wing had been designed, it was important to perform some calculations, based on empirical data, to obtain the approximate wing performance. These values could be used by the other teams and validated against CFD. The lift curve slope in the operational domain was obtained using a formula in Anderson for high aspect ratio wings, which is shown below [8].

$$a = \frac{a_0}{1 + a_0/\pi e_1 AR} \quad (8)$$

Where a is the wing lift curve slope, a_0 is the aerofoil lift curve slope (calculated as 5.95 from Xfoil) and e_1 is an efficiency factor that was taken to be $e_1 = 0.95$ [8]. Solving this equation led to $a = 5.85$.

5 Wing Design Conclusions

The aim of the wing design was to achieve an aerodynamically efficient wing, which is an elliptically distributed load. This has been achieved. The designed wing has a lift distribution that is only 2.61% different from a perfectly elliptical distribution. Therefore, it can be assumed that the design is the minimum drag design. However, once the design was passed to the structures team, they claimed that additional weight would be needed to support this configuration [9]. Therefore, the mass of the wing is perhaps higher than what is optimal. In future design iterations, the aerodynamics and structures team would work together in order to obtain a wing with the a similar efficiency, at a lower weight. This could involve an automated process. Unfortunately, this was not possible due to the time constraints of the project, but will be looked at in future. It may be desirable to revisit using the collocation method to design a wing with one taper ratio for the whole span, and look at using CFD in earlier stages of the design to obtain more accurate performance characteristics.

This section also highlighted the strong advantages of using inexpensive methods to design a wing. The wing geometry was needed early on in the project and without these methods, it would have been extremely difficult to design the most aerodynamically efficient wing in the time frame.

Part II

Propeller CFD Analysis

This section concerns using Star CCM+ to analyse the performance of the UAV's pusher propeller (used in the cruise phase of flight), and compares to initial BEM calculations. The CFD software that was used is Star CCM+, which is developed by Siemens, and is the industry standard software for modelling CFD simulations. The software solves the RANS equations for a given turbulence model. CFD was validated against BEM, and a mesh convergence study was undertaken. The method for these simulations was developed with inspiration from Austin Schenk [10], who has developed Star CCM+ simulations for propellers to compare to experimental data, making the method easier to validate. This method is similar (besides geometry) to that used in Shahriar's report [11] for rotor performance analysis.

The main objectives of the CFD simulations that will be discussed are as follows:

- Mesh Convergence Study.
- Compare CFD to BEM in high speed cruise and loiter.
- Analyse the effect of the fuselage in front of the propeller.
- Analyse the effect of the propeller wake on the tailplane.

6 Model Setup

6.1 Geometry

The propeller with hub geometry was imported and a disc, that just encapsulates the propeller, was formed around this. This disc formed the rotating mesh in the simulation. There was then

a larger cylinder that surrounded the disc and extending three diameters downstream. This was done to accurately capture behaviour of the wake. Surrounding this was a large bullet domain. The hemispherical side is the velocity inlet, and is the recommended inlet for such simulations by Star CCM+, as seen in figure 6. The flat face at the bottom is a pressure outlet, and the cylinder side wall is set to be the free stream, making this a virtual wind tunnel setup.

Once the parts were created, it was essential that they are tessellated extremely finely. As propellers have very intricate designs, it is essential that no errors arise from the creation of the parts. This practice was suggested by [10].

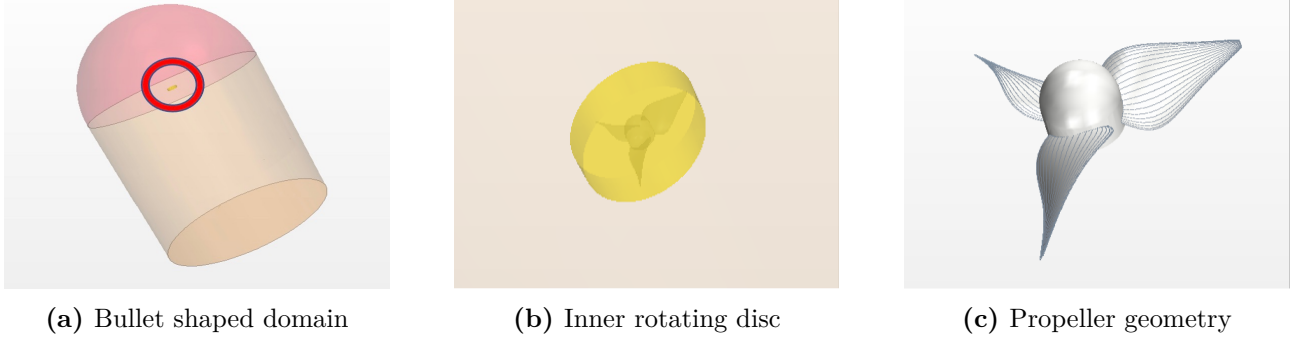


Figure 6. Geometry set up (no fuselage case)

6.2 Mesh

This simulation has two different meshes. The main fluid domain mesh, and the rotating mesh for the propeller. All of the mesh sizes are set relative to the base size. They are set such that the maximum surface size never exceeds 43% of the base size, and the minimum surface size is 1% for the fluid mesh, and 0.02% in the rotating mesh [10]. In addition, surface and volumetric controls were applied to enforce a finer mesh in the wake. In the fuselage case, an additional volumetric control was implemented as a cylinder so that a finer mesh could be generated near the surface of the fuselage. Example mesh cross sections are shown in figure 8. This resulting mesh had between 1.5-3 million elements (depending on fineness and whether the fuselage was being modelled).

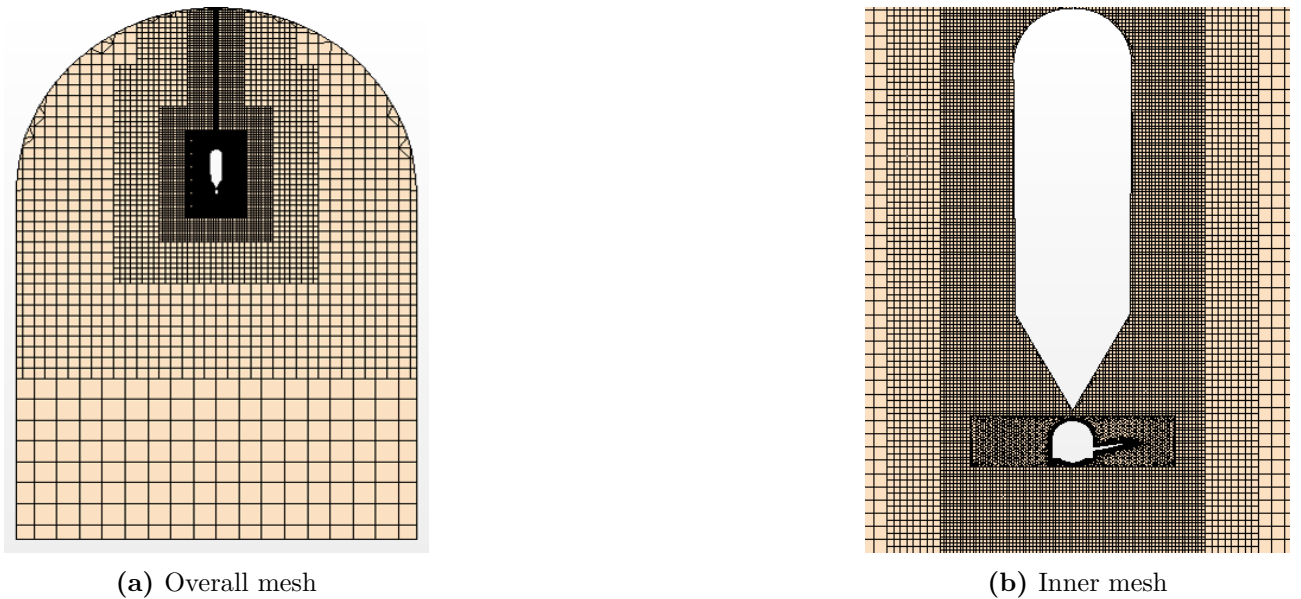


Figure 7. Mesh setup (fuselage case)

6.3 Physics Models

The selection of appropriate models is important to ensure the results were accurate, while not over-constraining and adding unnecessary computational complexity. These models were recommended by Star CCM+ for rotational flow, and were also used in [10].

Type	Model	Justification
Spatial	3D	Essential for propeller simulations as the model is 3D. Unfortunately there is no symmetry in this model so an axis-symmetric model could not be used.
Time Dependence	Implicit Unsteady	The propellers accelerate flow with respect to time, which is how they generate thrust. Therefore, it is essential to account for this with an unsteady model. An implicit model is required as the flow is turbulent and an explicit model would be unstable.
Coupling	Coupled	At the tips of the blades, $M \approx 0.5$, meaning that the flow is compressible. It is not appropriate to use the segregated flow model for a few reasons. The main one is that in compressible flows, the energy and momentum equations are coupled, this cannot be solved by a segregated flow solver.
State of Fluid	Ideal Gas	As the flow is compressible, it is not valid to assume a constant density. The ideal gas model allows for accurate modelling of the compressibility of the air.
Turbulence	$K - \omega$	The $K - \omega$ turbulence model is commonly used for sub-sonic flows. It is highly robust and more accurate for separated flows (this is proved in Max's report [2]). A reynolds stress model is more accurate when the flow is attached, however is much more computationally expensive.
Wall Treatment	All-y+	Accurately depicting the flow behaviour near the wall, before the log law is applicable [7]. Fine prism layers were implemented on the propeller surface to accurately capture this behaviour. This model is the standard wall treatment model for such a flow.

Table 6. CFD physics conditions

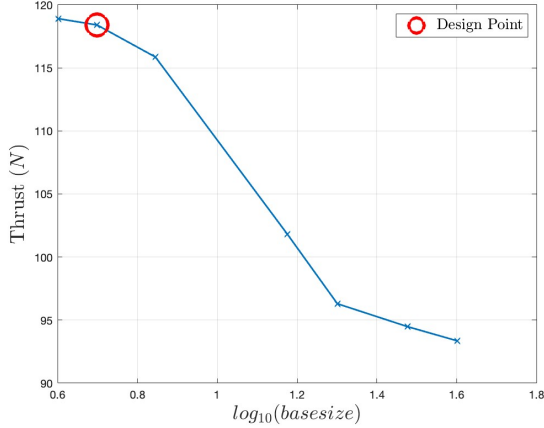
6.4 Running the Simulation

Due to the very large number of elements, simulations were run on high performance computers using sixteen cores. The simulation ran by performing five iterations at one time step. The propeller rotated by one time step and performed another five iterations. This process continued until the overall system converged (example residuals shown in appendix B). It usually took around 12 hours to fully converge.

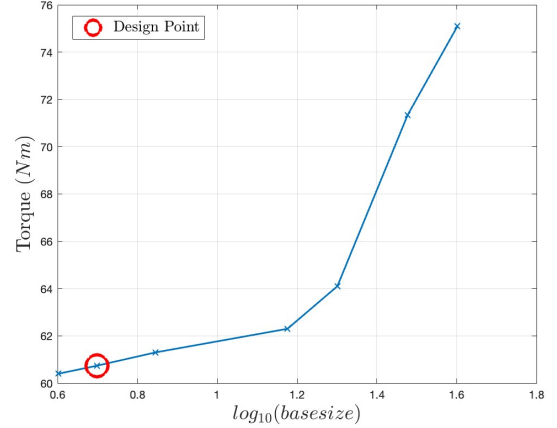
6.5 Mesh Convergence and Validation

6.5.1 Mesh Convergence

A mesh convergence study was conducted in order to verify at what point the solution has converged. In this case, this is especially important because saving time when running the simulations is critical when each simulation is so time consuming. In the study, the computational limit of the computer was reached. Making a finer mesh resulted in the computer crashing. This study was completed in the no fuselage case at 35 m/s free stream velocity, as this value was considered to be the loiter cruise speed earlier on in the design process.



(a) Thrust convergence



(b) Torque convergence

Figure 8. Mesh convergence and design points

A base size of 5 m was selected. This is because it is well within 3% error from a base size of 4 m, and there are around 500000 fewer elements, which saves substantial computational power and time.

6.5.2 Validation

This simulation was run for a stock propeller for which there is experimental data. An APC 10x7 propeller was used. The thrust and torque was compared to experimental data from the UIUC experimental database [12]. Although this propeller is very different from the pusher propeller (and much smaller), all propellers use the same physics models in Star-CCM+, so it was deemed appropriate to use as a validation, the CAD model, and inspiration, for this verification came from [10].

	CFD	Experimental	Error (%)
Thrust (N)	2.34	2.16	8.33
Torque (Nm)	0.0766	0.0740	3.51

Table 7. Comparison of experimental and computational results for an APC 10x7 propeller [12]

7 Propeller Results

Having verified the CFD method, simulations were run at different rotational speeds, in both the loiter cruise and high speed cruise, and this was then compared to the BEM method. Q Blade was used for this analysis, initially there was a lot of error between BEM and CFD, however, a 3D BEM correction was used, which reduced error in the results. Detailed analysis of BEM is mentioned in Parasdeep's report [3].

The Fuselage was modelled as a cutout in the domain. The shape of the fuselage itself was accurate to the design, however the wing was not included. Although the wing would almost certainly have had an effect on the thrust produced, it was deemed minimal. Figure 9 shows the true velocity distribution at the inlet of the propeller from the full aircraft simulation. The small dip in the free stream flow represents the effect of the wing. It only reduces the velocity by about 15% in a very small region, having a minimal impact on the overall velocity distribution. This means that it was appropriate to model just the fuselage in front of the propeller.

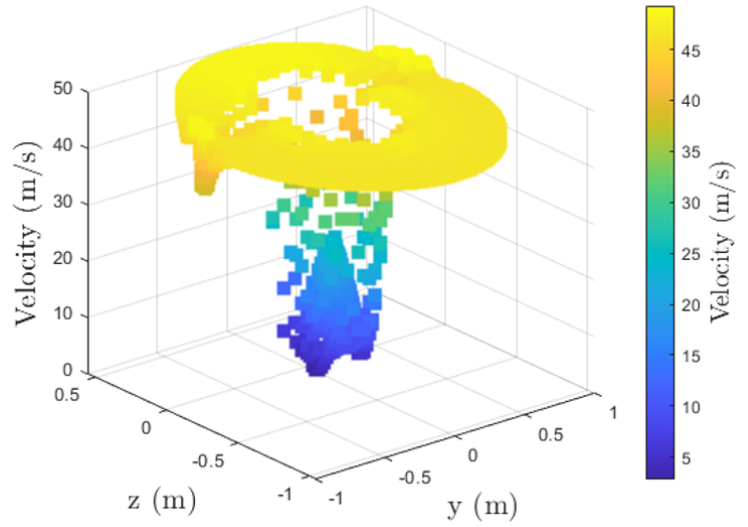


Figure 9. True velocity distribution at the propeller inlet for 41.1 m/s [2]

7.1 Loiter Cruise

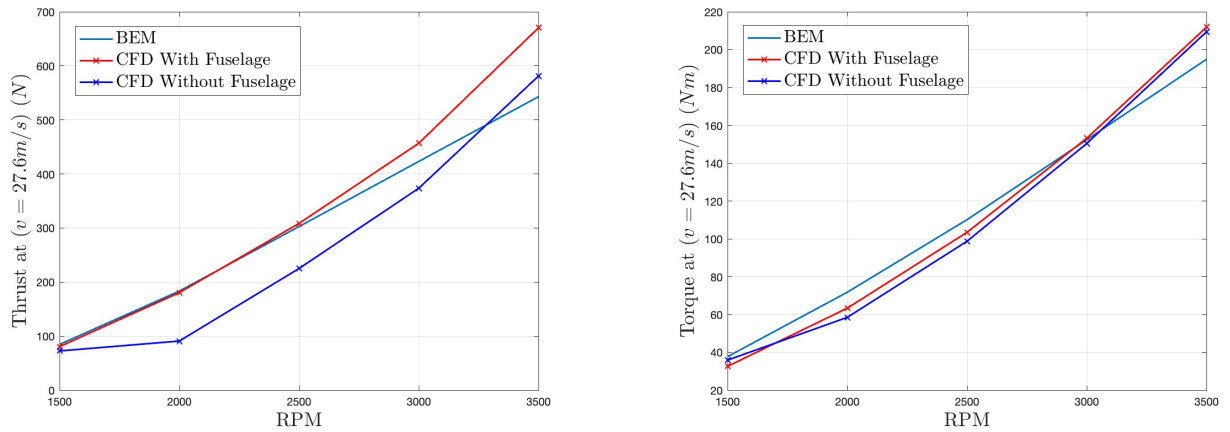


Figure 10. Thrust and Torque at 27.6 m/s

RPM	η_p			% error from BEM	
	BEM	CFD with fuselage	CFD without fuselage	Fuselage	No fuselage
1500	0.395	0.434	0.404	10.1	9.91
2000	0.337	0.375	0.234	11.5	39.2
2500	0.290	0.314	0.256	8.50	17.0
3000	0.244	0.262	0.227	7.39	10.6
3500	0.210	0.238	0.216	13.6	0.370

Table 8. Propulsive efficiency at 27.6 m/s

7.2 High Speed Cruise

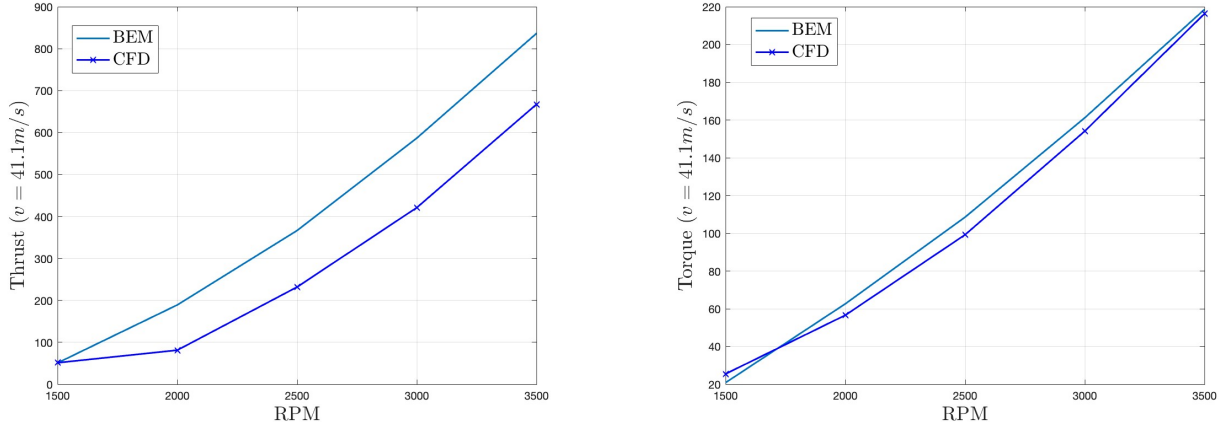


Figure 11. Thrust and Torque at 41.1 m/s

	η_p		% error from BEM
RPM	BEM	CFD without fuselage	No fuselage
1500	0.639	0.526	17.6
2000	0.592	0.282	52.5
2500	0.530	0.367	30.8
3000	0.476	0.358	25.0
2500	0.429	0.346	19.4

Table 9. Propulsive efficiency at 41.1 m/s

The equations used to obtain η_p are shown in appendix C. Due to time constraints no data was taken for the fuselage case at this speed. This is something that will be done in the near future.

7.3 Propeller Wake

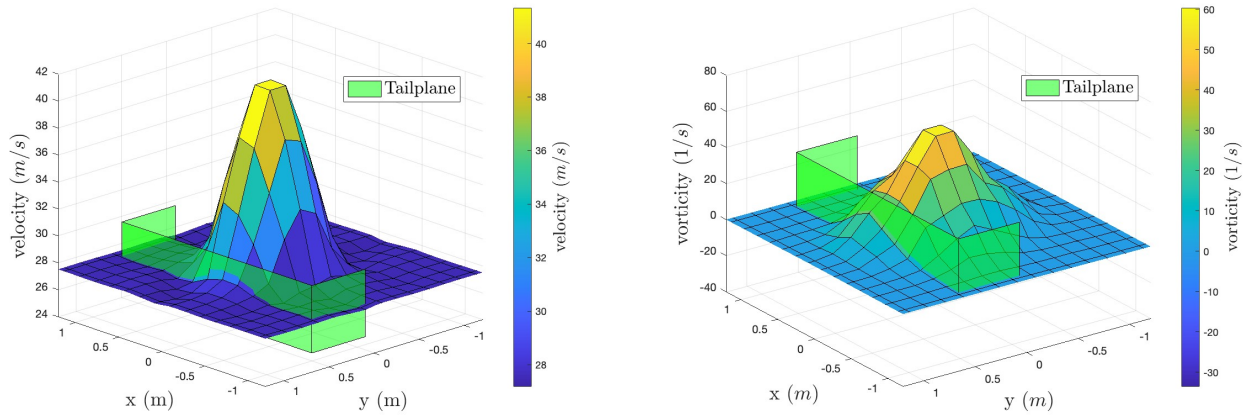


Figure 12. Velocity and vorticity at the leading edge of the tailplane caused by the propeller at 27.6 (m/s) at 2500 rpm

Figure 12 shows the velocity and vorticity distributions on the plane where the leading edge of the tailplane sits ($z = 4.4$ m). The tailplane is extruded to show the magnitude of the velocity and vorticity that is incident on the leading edge.

8 Discussion of Results

The results obtained show a good correlation with BEM, especially in torque, further validating the use of CFD for propellers. There are however some significant errors as CFD seems to almost always under-predict the thrust. There are a few reasons for this. Firstly, as seen in appendix B, the converged values of thrust and torque are oscillatory. This is because the simulation is unsteady so there is still some variation of thrust with time even after the simulation has converged. A mean value of this convergence was taken over 1 time period, however there could potentially be a small error from reading the data.

Another cause of low thrust predictions is a drop in density in the propeller blade wake. The reason for this drop is probably due to the effect of the wake, or added mass effects [13]. This drop in density is shown in figure 13. More simulations will be performed at different conditions to fully investigate and understand the theory behind this. At both airspeeds, the thrust was particularly different from BEM at 2000 rpm, this is something that needs investigating by running simulations at more rotational speeds, as thrust increases rapidly beyond 2000 RPM.

It is shown in [14] that turbulence can lead to an increase in propulsive efficiency of a propeller. Their results show that η_p increases by about 15% when the flow is turbulent compared to laminar at high advance ratios, and they suggest that: *'turbulence ingestion increases generated thrust due to the increase in pressure by the propeller blade to compensate for the drop of pressure induced by the incoming turbulence'* [14]. There are a lot of theories that can explain the discrepancies between BEM and CFD, however more simulations, and experiments, need to be conducted to accurately explain these differences. This research could have long term benefits for increasing propulsive efficiency of propellers, however for the purposes of designing a UAV, the CFD values can be used as they have been validated against BEM and experimental data, and as CFD uses more complicated models than BEM, it is expected to be a more accurate model.

A particularly interesting phenomenon that could not have been observed if it was not for CFD is the increased performance of the propeller with the fuselage in front of it. Initially, this result came as a large surprise, as it was expected that the blanketing affect of the fuselage would cause the propeller to produce less thrust. From comparing the BEM results in tables 8 and 9, the propulsive efficiency in the high speed cruise is significantly higher than for the loiter cruise. This implies that the increased free stream velocity is the cause. The fuselage positioning is accelerating the flow due to the high pressure at the front of the fuselage, inducing a favourable pressure gradient. There is flow separation towards the end of the fuselage, where it tapers away, however the overall consequence of the fuselage is a faster flow field at the propeller inlet. Another consequence of the fuselage is that the propeller itself produces less drag (as the majority of it is taken by the fuselage), further increasing it's efficiency as the thrust force calculated in Star CCM+ is $T - D$. Although the flow speed at the root of the blades is significantly less with the fuselage (due to flow separation from the fuselage see

	Free stream	Wake
Velocity (m/s)	27.6	29.1
Vorticity ($1/s$)	0	18.6

Table 10. Maximum velocity and vorticity at the tailplane leading edge

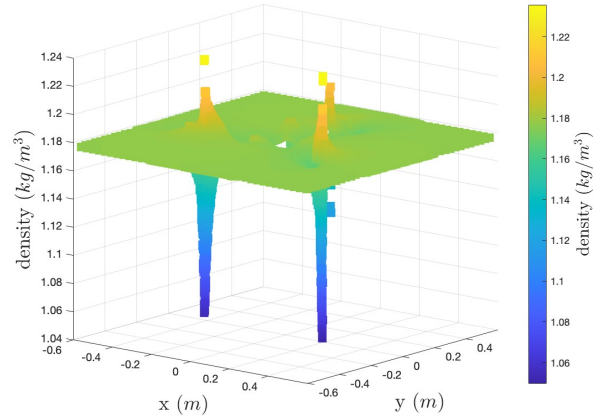


Figure 13. Density variation on the plane of the propeller

figure 24), this section of the propeller will produce less thrust compared to the tip regardless as the tangential velocity is much smaller at the root. The fuselage will also make the inlet airflow smoother and more uniform, further increasing efficiency. In Appendix B, some raw CFD data is shown. The oscillations in the steady state values are significantly less with the fuselage compared to without it. This suggests that the flow is more steady, and therefore more efficient as an inlet [10]. This effect is used in many modern aircraft, such as the Honda jet [15], to achieve a higher engine efficiency. More studies need to be conducted to further investigate and understand this phenomenon, and see how the fuselage design can be refined to further increase efficiency.

Table 10 shows the maximum values of the velocity and vorticity that the horizontal tailplane will experience in the loiter cruise. The increase in velocity is less than 10%, so is unlikely to be of any significant detriment to the performance. The non dimensional vorticity is defined as $N_\omega = |\omega|c_h/U_\infty$ [16]. Using $c_h = 0.461m$ from [17], this leads to $N_\omega = 0.31$. This number is quite small, and is unlikely to have a significant impact on tailplane performance. This number usually needs to be around 1 to have a significant impact [16]. However, at high angles of attack, the impact of the wake may be quite significant as the tailplane may become incident on the peak of the vorticity distribution shown in figure 12, so should be analysed further at such angles. For the cruise phase, which is the vast majority of the flight, it will have a very limited impact on performance and it's impact can be neglected.

9 Propeller CFD Conclusions

The CFD simulations were extremely successful because they provided more accurate thrust and torque curves, and it highlighted available opportunities to improve propeller efficiency. Firstly, the results showed a particularly good correlation with BEM for torque, and a reasonable correlation for thrust. The BEM method has many limitations, mentioned in Parasdeep's report [3], so is not a very accurate representation of reality. As CFD uses far more advanced models and has been validated extensively, it is sensible to accept it's results as closest to reality. The truest reality of the propeller's performance would come from flight testing and experiments. CFD allowed more accurate torque and thrust distributions to be obtained. This was useful as it enabled the propulsion team to refine their engine choice to better reflect reality. There were some cases where there were very significant discrepancies between CFD and BEM, and the error sometimes exceeded 50%. These situations need to be thoroughly investigated, with simulations being re-run, to ensure they are correct. Unfortunately, due to the time constraints of the project, and the duration of each simulation, there was not sufficient time to run more tests. In future, significantly more CFD data points will be added, to verify the results further and to eliminate any anomalous results.

The reasons for the larger disagreements in the thrust readings are not completely clear. The main theories presented in this report are: drops in density, propeller hub drag and turbulence; however, simulations and experiments need to be run at different conditions to determine what is actually causing this disparity in the results.

It seems that the propeller wake will have an insignificant impact on the performance of the tailplane. This is due to the low magnitude of the wake vorticity calculated by CFD. It would be desirable to conduct further investigations by further refining the wake mesh to get more accurate results, and by changing the angle of attack of the propeller relative to the free stream, to see if that causes a higher magnitude of vorticity to come into contact with the tailplane.

The most exciting conclusion from the CFD is that: by having a pusher propeller configuration, propulsive efficiency can be increased without making any significant sacrifices to the fuselage design. This has the potential to reduce fuel consumption significantly, saving weight, and therefore reducing operating costs. The results in table 8 show that η_p is around 10% higher when the fuselage is in front of the propeller. In order to further investigate, and validate, these results, more simulations have to be performed at different free stream velocities. It would also be beneficial to see how changes to the fuselage design will impact the thrust. Various parameters of the fuselage, such as length, diameter and fuselage taper angle should also be altered to investigate the impact that this has on efficiency. This could lead to significant improvements in propeller efficiency for very little cost.

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Appendix

A Collocation Method Code

```
1 %This function just runs collocation method for each chord and
2 %twist distribution
3
4 function [Cl,Cd_i,LD,error] = collocation(taper,a0,alpha_0,alpha_t)
5
6 %From brief and Raymer
7 AR = 8.5;
8 b = 6;
9 S_ref = b ^ 2 / AR;
10
11
12 %dimensions
13
14 theta1 = pi/5;
15 theta2 = 2*pi/5;
16 theta = [theta1,theta2];
17
18 %taper = 0.75;
19 %get distributions for chord and twist
20 c_root = S_ref./(0.5.*(1+taper).*b);
21
22 c_tip = taper.*c_root;
23
24 c = c_root * (1 - (1 - taper)*cos(theta));
25
26 %gets twist distribution in terms of theta, as this is written as
27 %y=scos(theta)
28
29
30 alpha = alpha_0 * (1 - ((alpha_0-alpha_t)/alpha_0)*cos(theta));
31 alpha = alpha .* pi ./ 180;
32
33
34 %4 not 8 as b is semi span
35 mu = a0 .* c ./ (4*b);
36
37 %solve collocation code
38 syms A1 A3
39
40 eqn1 = mu(1)*alpha(1)*sin(theta1) == A1*sin(theta1)*(mu(1)+sin(theta1)) + A3*sin(3*theta1)
41        *(3*mu(1)+sin(theta1));
42 eqn2 = mu(2)*alpha(2)*sin(theta2) == A1*sin(theta2)*(mu(2)+sin(theta2)) + A3*sin(3*theta2)
43        *(3*mu(2)+sin(theta2));
44
45 %Calculate lift and drag coefficients and error
46
47 [A1,A3] = solve(eqn1,eqn2,A1,A3);
48 A1 = double(A1);
49 A3 = double(A3);
50 %multiply by 2 for the whole wing
51 Cl = 2*pi*AR*A1;
52 Cd_i = 2*pi*AR*((A1^2)+3*(A3^2));
53 LD = Cl/(Cd_i);
54 error = abs(A3/A1);
```

Listing 1. Collocation function, performs collocation method for each chord and twist

```
1 function [minerror,alpha_0min,tapermin] = runner(alpha_t,a0)
```

```

2 %iterates collocation method for different alpha_0 anf taper ratios and
3 %then gets the minimum error and the values corresponding to it
4
5
6 alpha_0 = [1:0.1:4];
7
8 taper = linspace(0.3,1,length(alpha_0));
9 for i = 1:length(taper)
10     for j = 1:length(alpha_0)
11         [Cl(i,j),Cd_i(i,j),LD(i,j),error(i,j)] = collocation(taper(i),a0,alpha_0(j),
12             alpha_t);
13     end
14 end
15 %%
16 [X,Y] = meshgrid(taper,alpha_0);
17 minerror = [];
18 for i = 1:length(X(1,:))
19     a = error(i,:);
20     [minerror(i),idx] = min(a);
21     tapermin(i) = X(1,i);
22     alpha_0min(i) = Y(idx,i);
23 end

```

Listing 2. Runner function, iterates between different values of λ and α_r and obtains distributions of lowest error

```

1 clear
2 clc
3 %iterates the runner code for different tip alpha and gets the relavent
4 %values from that by running collocation again then plots it
5 a0 = 5.85; %Approximate value
6 alpha_t = [1:0.5:2]; %Tip angles of attack that will be iterated
7 hold on
8 for i = 1:length(alpha_t)
9     [minerror,alpha_0,taper] = runner(alpha_t(i),a0);
10    [a1,idx1] = min(minerror);
11    plot3(taper,alpha_0,minerror,'LineWidth',1.5)
12    [Clmin(i),Cdmin(i),LDmin(i),errorminmin(i)] = collocation(taper(idx1),a0,alpha_0(idx1
13        ),alpha_t(i));
14    taperminmin(i) = taper(idx1);
15    alpha_0minmin(i) = alpha_0(idx1);
16 end
17 grid on
18 xlabel('Taper Ratio','FontSize',18,'LineWidth',1.5)
19 ylabel('Root angle of attack','FontSize',18,'LineWidth',1.5)
20 zlabel('Error from elliptical','FontSize',18,'LineWidth',1.5)
21 legend('alpha_t = 1','alpha_t = 1.5','alpha_t = 2','FontSize',14,'LineWidth',1.5)
22 hold off

```

Listing 3. Runner2 function, iterates between different values of α_t and plots the line of minimum error from elliptical

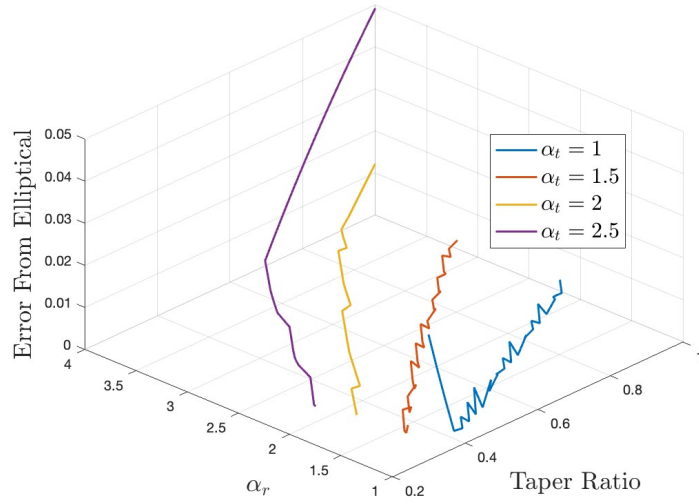


Figure 14. Most efficient wing designs from collocation

B CFD Raw Data

B.1 APC 10x7

The results for the validation case (APC 10x7 propeller) shown in 7 are shown below.

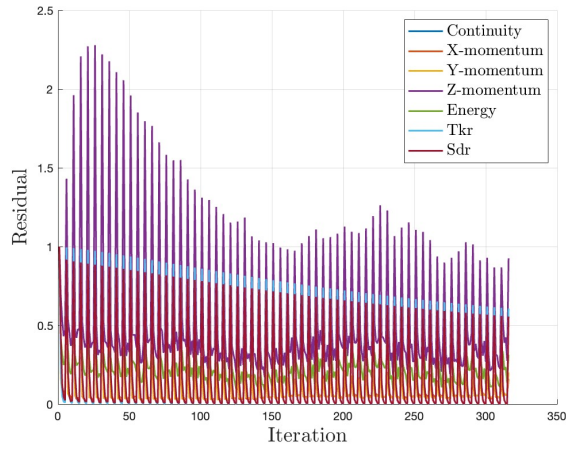


Figure 15. Residuals APC

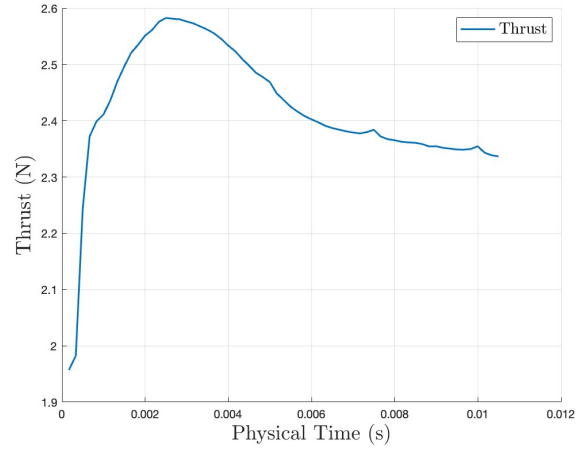


Figure 16. Thrust APC

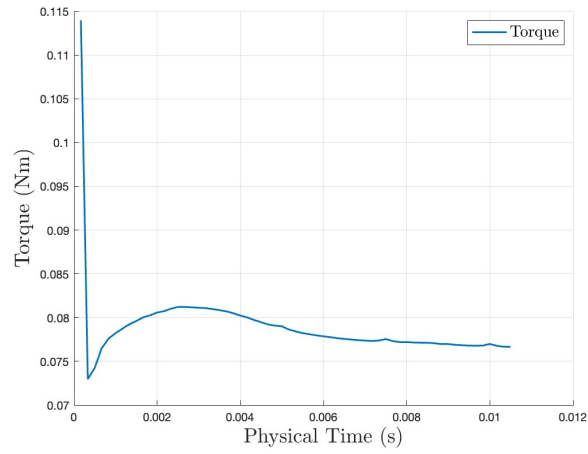


Figure 17. Torque APC

B.2 Raw data at 27.6 m/s and 3500 RPM

The aim of presenting this data is to show how thrust and torque still oscillated once converged, this is because of the unsteady models used. An average was taken.

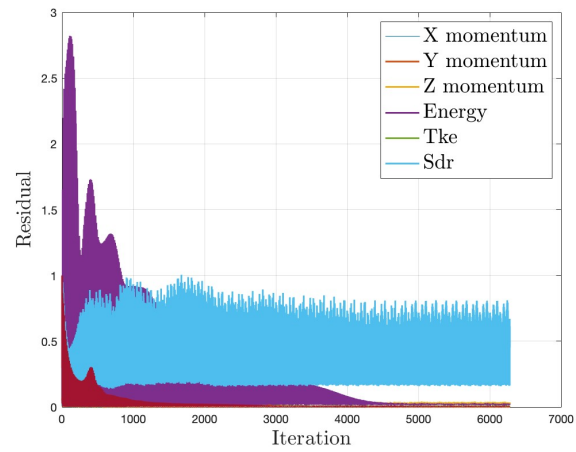


Figure 18. Residuals 3500 RPM

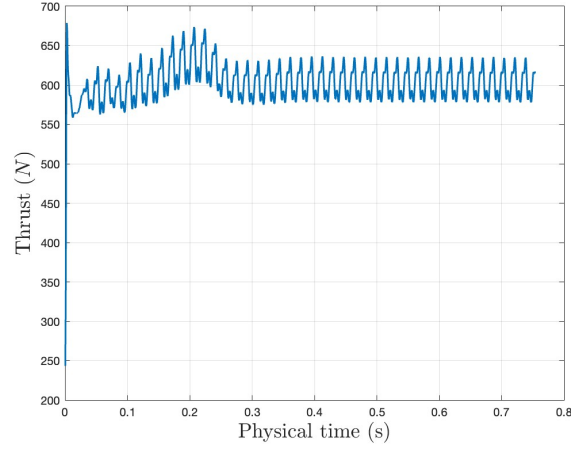


Figure 19. Thrust 3500 RPM

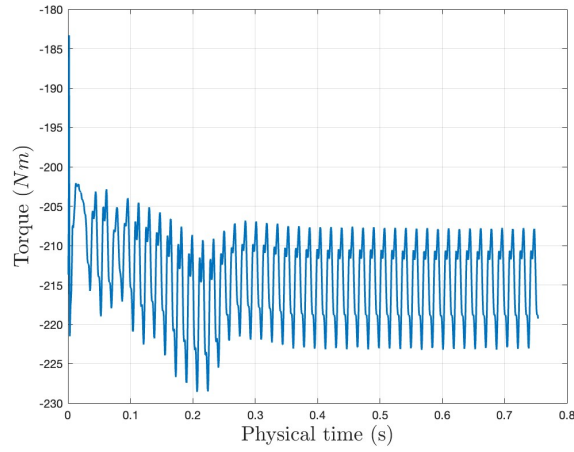


Figure 20. Torque 3500 RPM

B.3 Raw data at 27.6 m/s and 2000 RPM with the fuselage

The aim of presenting this data shows how the fuselage reduces oscillations in steady state torque and thrust, possibly linking to increased efficiency.

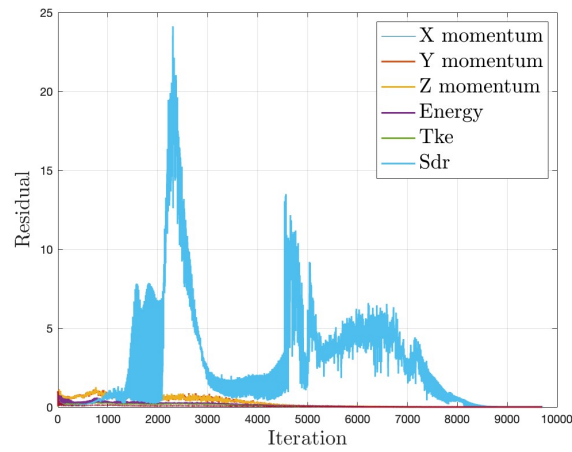


Figure 21. Residuals at 2000 RPM with the fuselage

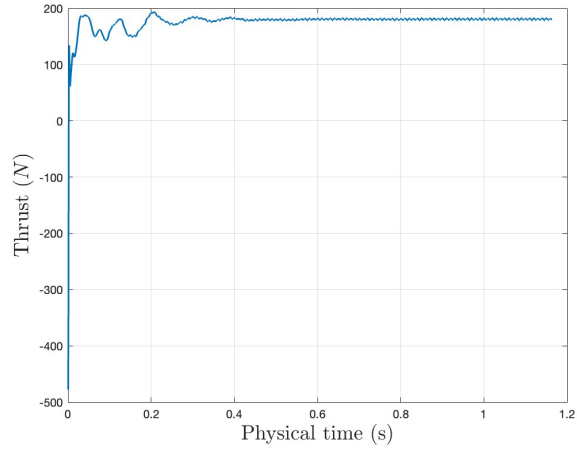


Figure 22. Thrust at 2000 RPM with the fuselage

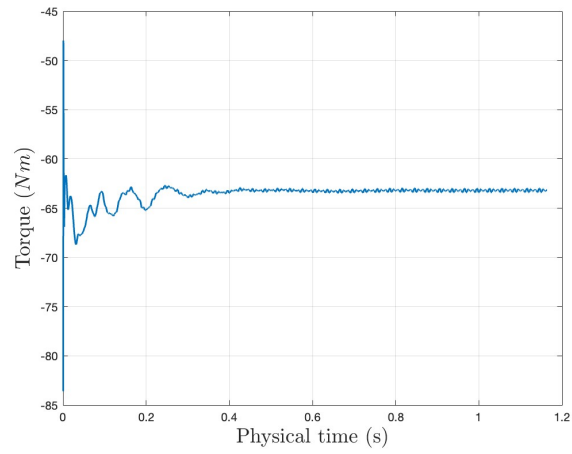


Figure 23. Torque at 2000 RPM with the fuselage

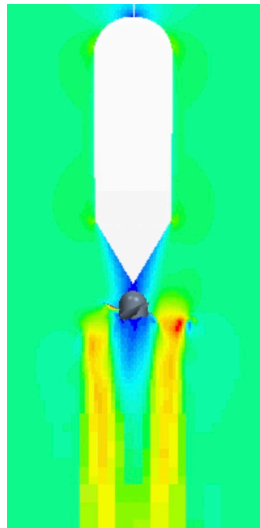


Figure 24. Velocity scene of the fuselage

C Propulsive efficiency

Propulsive efficiency is defined as:

$$\eta_p = J \frac{C_T}{C_P} \quad (9)$$

Where the J , C_T and C_P are defined as:

$$J = \frac{V}{nD} \quad C_T = \frac{T}{\rho n^2 D^4} \quad C_P = \frac{P}{\rho n^3 D^5} \quad (10)$$

n is the rotational speed in revolutions per seconds, and D is the propeller diameter. Power is calculated as $P = Q\omega$, where ω is rotational speed in radians per second [3].